Matrix inequalities and norms

Marcus's inequality for the permanent of block permanents

Difficulty: extreme **Importance:** interesting to specialist

Status: explicit conjectured inequality; no later resolution located

For a square matrix C of order r, write

$$\operatorname{per} C = \sum_{\sigma \in S_r} \prod_{i=1}^{r} c_{i,\sigma(i)}.$$

Let $m, k \geq 2$ be arbitrary integers and let $A \in \mathbb{C}^{mk \times mk}$ be Hermitian positive semidefinite. Partition A into an $m \times m$ array of $k \times k$ blocks A_{ij}, and form

$$G = (g_{ij})_{i,j=1}^{m}, \qquad g_{ij} = \operatorname{per}(A_{ij}).$$

Must

$$\operatorname{per} A \geq \operatorname{per} G$$

always hold?

Relevance

This asks how a costly matrix polynomial behaves under block aggregation by the same polynomial. It is a precise constraint on hierarchical permanent computation for PSD matrices; positivity is assumed for the whole matrix, not separately for off-diagonal blocks.

References

- I. M. Wanless, *Lieb's permanental dominance conjecture* (2022), Conjecture 3 and the following paragraph (primary manuscript).
- F. Zhang, *An update on a few permanent conjectures*, Special Matrices 4 (2016), 305–316, discussion of Marcus's permanent-of-permanents conjecture (primary manuscript; journal).
- E. H. Lieb, *Proofs of some Conjectures on Permanents*, Journal of Mathematics and Mechanics 16 (1966), 127–134, the two-block inequality (publisher's first page).

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Wanless records this inequality as open and identifies Lieb's proof for $m = 2$. Searches for "Marcus conjecture", "permanent of permanents", "block permanents", "proof", and 2025–2026 found no general resolution. This entry is specifically the inequality component: some historical formulations additionally conjecture an equality characterization. The trivial block sizes $m = 1$ and $k = 1$ are omitted, and no unqualified equality characterization is imported. Lieb's general permanental dominance conjecture would imply this assertion, but this is a separately stated historical conjecture.